UNIT-2

DATA LINK LAYER

The Data Link Layer is the 2nd layer of the OSI model. just above the Physical Layer

The main role of this layer to ensure reliable transfer between two directly connected nodes.

Need of data link layer.

- Physical layer only sends raw bits (0s and 1s)
- Errors, loss, duplication can happen.
- No structure or reliability.

DLL solves this by :-

- Converting bits into frames.
- Detecting and correcting errors.
- Controlling data flow.
- Managing communication between directly connected nodes.

Services Provided by Data Link Layer (DLL).

① Flow Control.
② Framing.
③ Error control. IMP

FLOW CONTROL 2022

Flow control is a technique used to control the rate of data transmission between sender and receiver so that:-

Sender does not send data faster than the receiver can handle.

Need of Flow control.

- Imagine. sender is fast and Receiver is slow.

→ Without flow control

- Receiver buffer gets overflowed.
- Data gets lost.

→ So, Flow control ensures:-

- Smooth data transfer
- No data loss.
- Efficient communication.

There are two main methods

① **Stop - and - wait flow control** 2022

- In this sender sends one frame and wait for ACK (acknowledgement).
- After receiving ACK → sends next frame

→ It is very simple but very slow.
Easy to implement.

Example → Like sending one message and waiting for 'OK' before sending the next

② **Sliding window flow control.**

- In this sender can send multiple frames before receiving ACK.
- Uses a window size (W).

Working

1. Sender sends multiple frames (up to window size).
2. Receiver sends ACKs
3. Window slides forward.

Example

- Window size = 3
  Sender sends: Frame 1, 2, 3 (without waiting)
- After Ack of Frames 1 → window moves → send frame 4.

★ It is efficient and high data transmission speed.

★ But more complex and needs buffer memory.

FRAMING IMP

Framing is the process by which the Data link Layer divides a continuous stream of bits into manageable units called frames.

It is the first and most important function of the Data Link Layer.

Need of framing-

When data comes from the Network Layer

- It is a continuous stream of bits
- No clear boundaries.

→ without framing

- Receiver cannot detect
  - Start of data
  - End of data.
- Error detection becomes difficult.

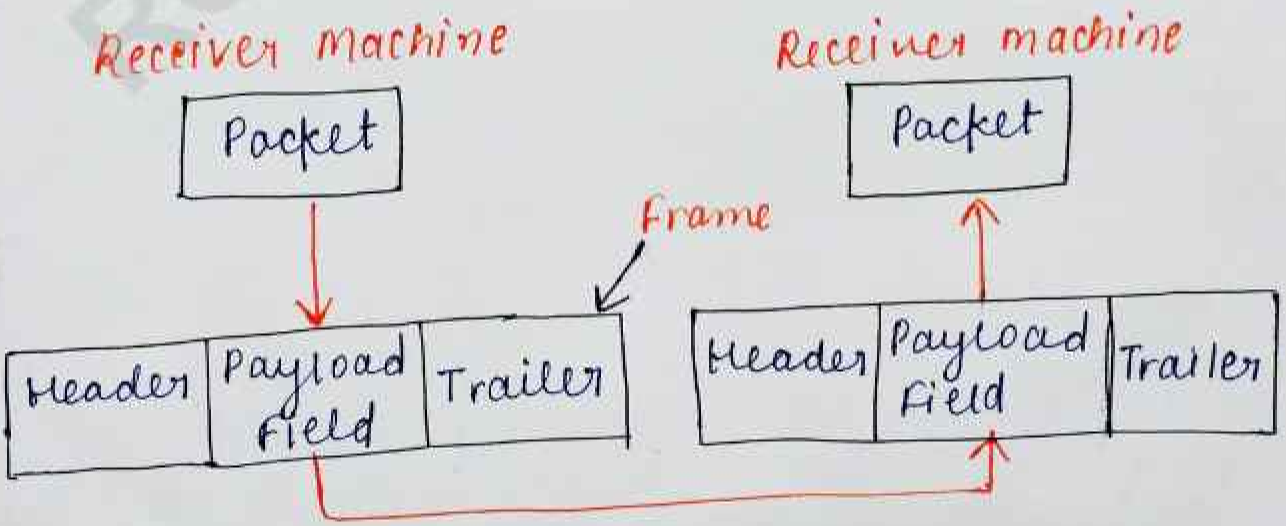
Synchronization fails.

What framing Provides.

Framing helps in:-

1. Data Organization - Breaks data into struct-ured units.
2. Synchronization - Sender and receiver stay aligned.
3. Error Detection - Each frame has error-checking bits.

Structure of a frame.



1. Header.

It contains. → • Source MAC address
• Destination MAC address
• Control Information
• Sequence Information.

2. Payload (Data)

Actual data from Network layer.

3. Trailer.

Error detection bits (CRC).

Problem in Framing:-

How does receiver know frame boundaries?

This is called:

Frame Delimitation Problem.

Solution → Use different framing techniques

Techniques of framing.

There are 4 main techniques:-

① Character count. → The first field of the frame specifies the length (number of bytes) in the frame.

Format → | Length | Data |

Example → 5 HELLO.

Means- frame contains 5 characters: HELLO

Problem → If length field gets corrupted, the receiver losses synchronization.

- Entire frame becomes invalid

→ That's why it is not reliable.

(2) Byte Stuffing (Character - Oriented framing)

It uses special character.

- FLAG → marks start and end
- ESC → escape character.

Format → FLAG | Data | FLAG.

Problem → If FLAG appears inside data, it create confusion.

Solution → Add ESC before FLAG.

(3) Bit Stuffing

- In this Frame start and ends with Flag pattern. → 01111110
- Used in Bit - oriented protocols (like HDLC)

Problem - If same pattern appears in data, it create confusion.

solution - Insert 0 after every 5 consecutive 1s.

☆ It is very reliable, no ambiguity and widely used.

☆ Slight increase in data size.

④ Physical Layer Coding Violations.

- Uses special signal patterns (not used in normal data).

→ These patterns mark:
  - Start of frame
  - End of frame

Example - Special voltage levels or signal transitions

Advantages	Disadvantages.
• No extra bits needed	• Depends on encoding scheme
• Efficient	• limited use

ERROR CONTROL

- Error Control is the process of detecting and correcting errors that occur during data transmission.

- When data travels through a network, it may get corrupted because of noise, interference, or signal loss. Error control ensures reliable communication.

- Error control in the data link layer is based on ARQ (automatic repeat request), which is the retransmission of data.

- ARQ → Automatic Repeat Request (ARQ)
  If receiver detects error, it requests sender to resend data.

Types of ARQ

1. Stop - and - wait ARQ
2. Go - Back - N ARQ
3. Selective Repeat ARQ.

Need of Error control

- Reliable data Transfer
- Prevent data Corruption
- Improve communication quality
- Maintain accuracy in network.

Example > If sender sends 101101. but receiver gets 101001 due to noise. error control techniques detect /correct this mistake.

Data Link Layer Protocol.

A Data link Layer Protocol is a set of rules that governs how data is transferred between two directly connected devices. It ensures that data is framed. transmitted. error-checked. and acknowledged properly.

Main functions.

1. Framing - Breaks raw bits into structured units called frames.
2. Physical Addressing Uses MAC (Media Access control) addresses to identify devices.
3. Error Detection & Correction - Detects errors using techniques like CRC (cyclic Redundancy check).

(4) Flow control - Ensures sender does not overwhelm receiver

(5) Access Control - Decides which devices get access to the shared medium.

Elementary Data Link Protocol

Elementary Data Link Protocols are the basic protocols used in the Data Link Layer of a computer network.

They define simple rules for ~~dat~~ sending data frames between sender and receiver over a communication channel.

Main Focus on :-

- Flow control - controls speed of data transfer
- Error control - handles lost or damaged frames
- Reliable delivery - ensures correct data reaches receiver.

Types

(1) Unrestricted Simplex Protocol.

- Data Flow only in one direction (sender → receiver.
- Receiver is always ready to receive data.
- No error checking or flow control.
- Used in ideal communication channel.

② Simplex Stop-and-wait Protocol

- Data flows in one direction only.
- Sender sends one frame at a time.
- After sending one frame, sender waits for acknowledgment (ACK)
- Prevents receiver overload.

Advantage - Simple and reliable

③ Simplex Protocol for Noisy Channel.

Used when channel may contain errors.

Frames may be lost or damaged

Uses :- Acknowledgment (ACK)

Negative Acknowledgement (NAK)

Retransmission of lost frames.

Advantages.

More Reliable in real networks

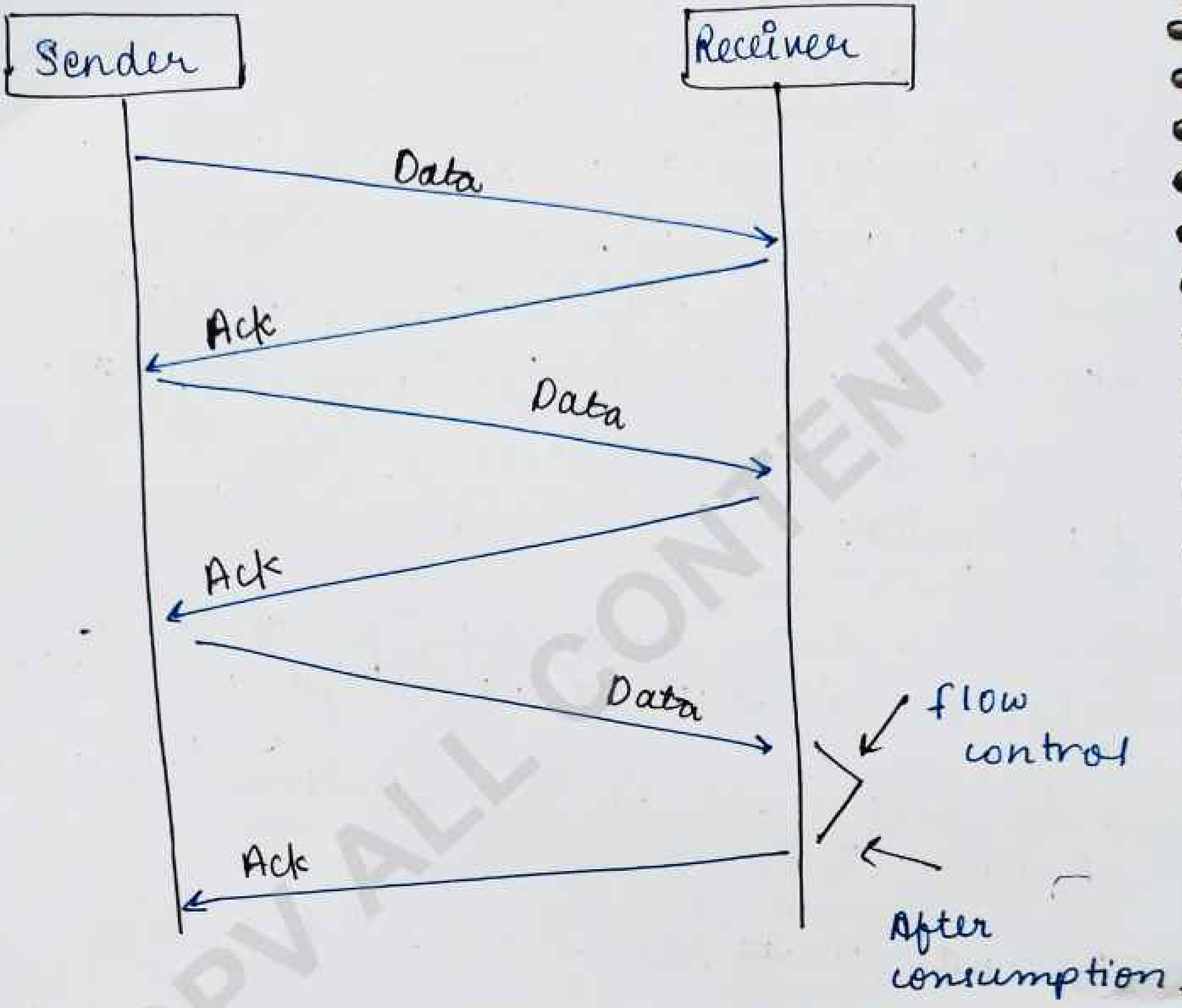
Sliding Window Protocol. 2023

Sliding Window Protocol is a flow control and error control technique used in computer networks. It allows the sender to send multiple frames before receiving acknowledgement, which improves transmissi-on speed and efficiency.

Main points

These are data link layer protocols for reliable and sequential delivery of data frame

- It is also used in Transmission control Protocol (TCP)
- It sending multiple frames at a time to the receiver.
- Each frame has sent from the sequence number.
- Sequence number are used to find the missing data in the receiver end.
- It's main purpose is to avoid duplicate data, so it uses the sequence number.



Types of Sliding Window Protocol. ARQ

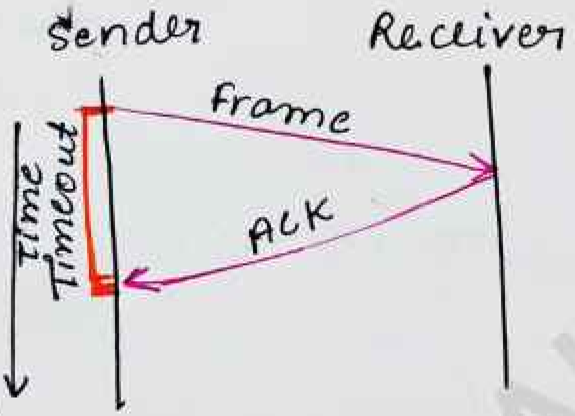
1. 1-bit or Stop-And-Wait protocol.
2. Go-Back N.
3. Selective Repeat.

STOP - AND - WAIT ARQ. Protocol. 2024

- After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.

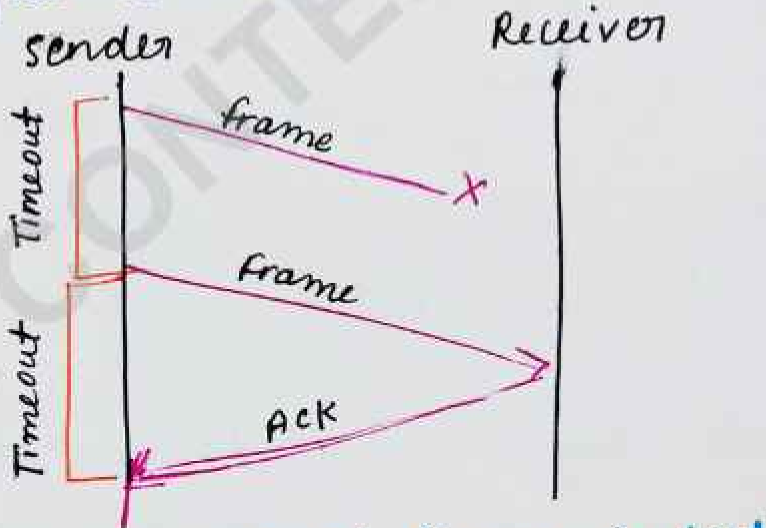
- If the acknowledgement does not arrive after a certain period of time, the sender times out and retransmits the original frame

Stop - and - wait ARQ → Stop and wait + Timeout Timer + sequence number



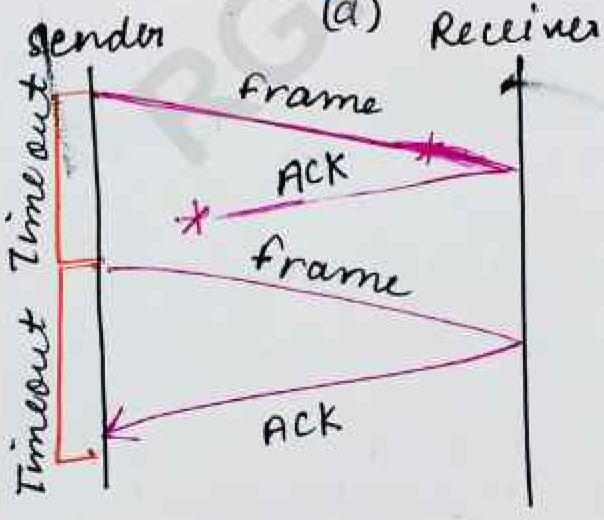
The ACK is received before the timer expires.

(a)



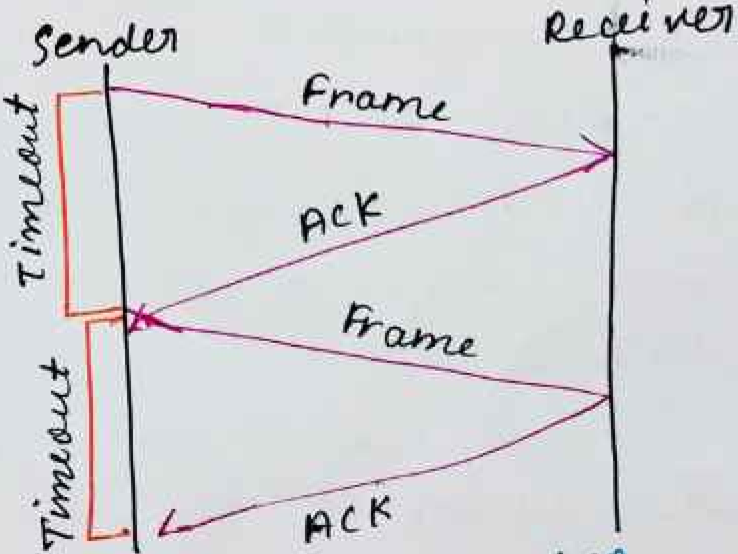
The original frame is lost

(b)



The ACK is lost

(c)



The timeout fires too soon.

(d)

Go - BACK - N ARQ. 2024 2025

'N' is the sender window size

→ Go - Back - N ARQ uses the concepts of protocol pipelining i.e. the sender can send multiple frames before receiving the acknowledgement for the first frame.

→ There are finite number of frames and the framed are numbered in a sequential manner.

→ The number of frames that can be sent depends on the window size of the sender

→ If the acknowledgment of a frame is not received within an agreed upon time period, all frames in the current window are transmitted

Go - Back - N ARQ.

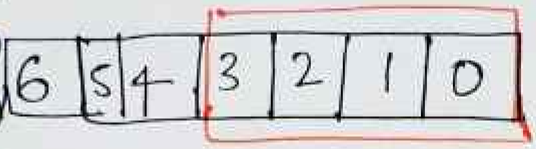
★ N - Sender's Window Size

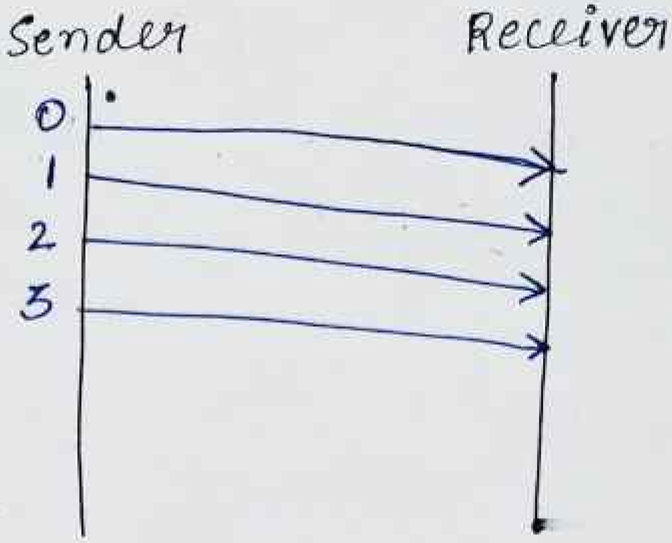
★ For example → if the sending window size is 4 (2^2) then the sequence numbers will be 0, 1, 2, 3, 0, 1, 2, 3, 0, 1 and so on.

★ The number of bits in the sequence number is 2 to generate the binary sequence 00, 01, 10, 11.

Go-Back-N ARQ.

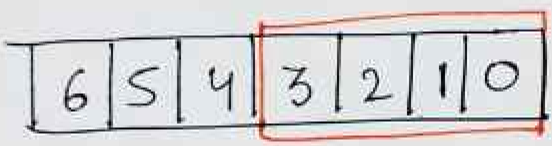
Step 1

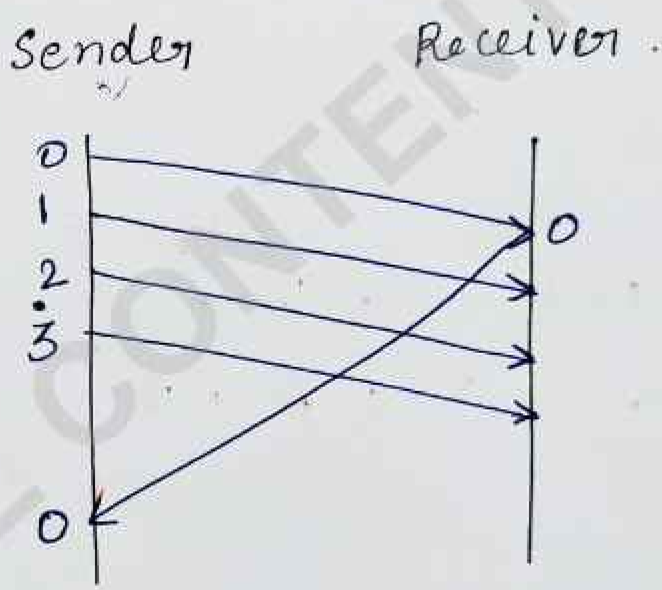




Window size = 4

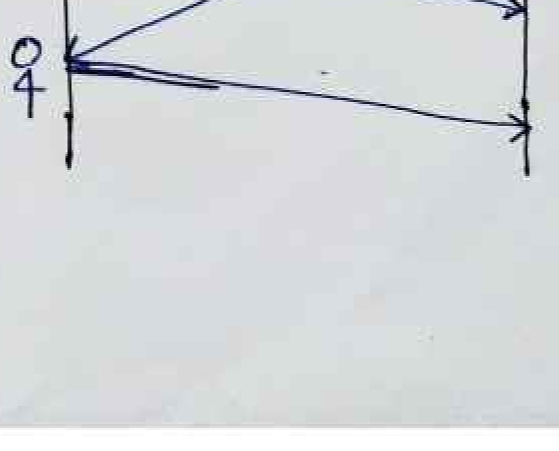
Step 2





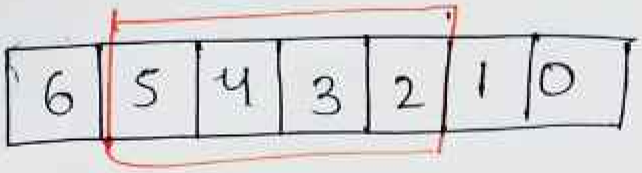
window size = 4

Step 3.

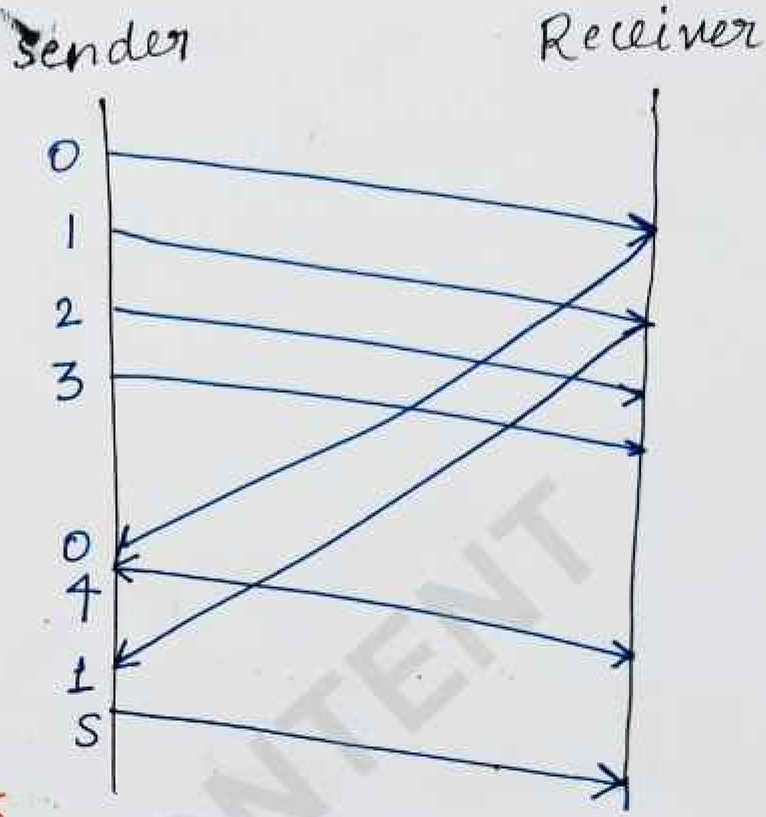


window size = 4

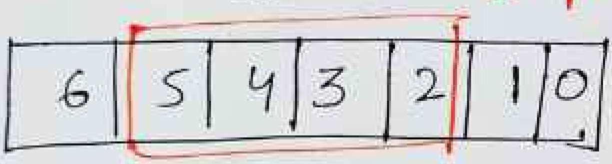
Step-4



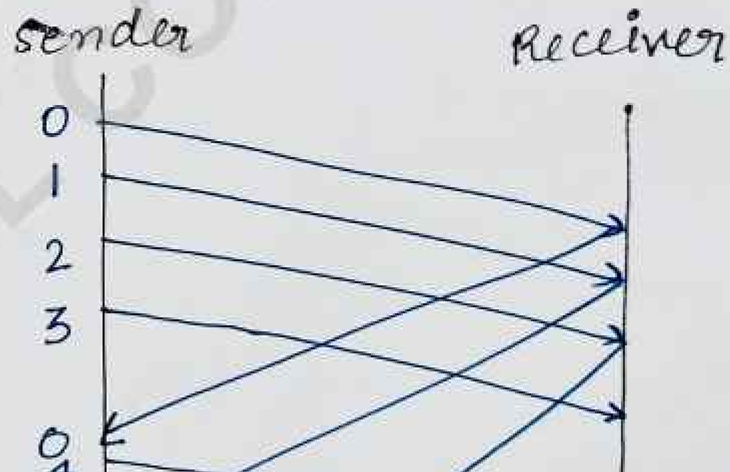
Window size = 4



Step-5



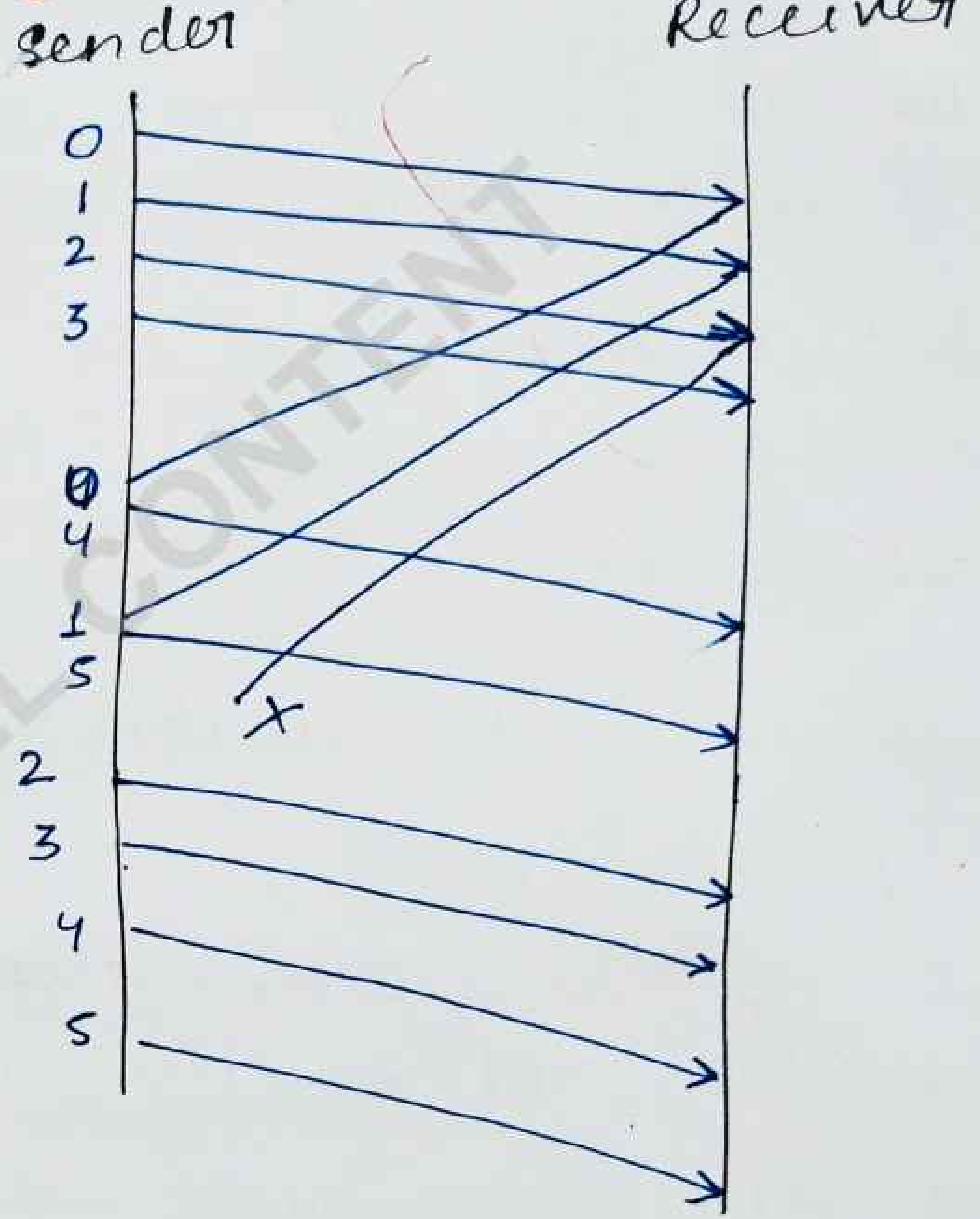
Window size = 4



Step 6

6	5	4	3	2	1	0

window size=4

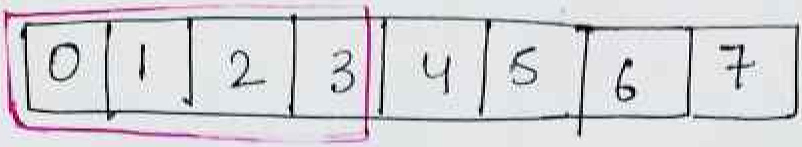


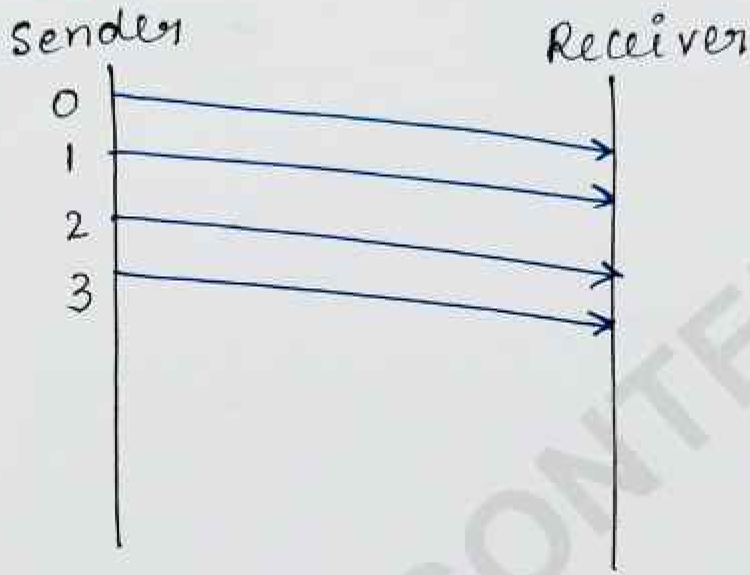
Selective - Repeat ARQ Protocol. 2024

- The Go - back - N ARQ protocol works well if it has fewer errors.
- But if there is a lot of error in frame & sending frames again. So we use Selective Repeat ARQ.
- It stands for Selective Repeat Automatic Repeat Request.
- If the receiver receives a corrupt frame, it does not directly discard it.
- It sends a negative acknowledgement to the sender.
- The sender sends that only frame again as soon as on the receiving negative acknowledgement.
- It provide efficient data transmission & fastest recovery

Step 1.

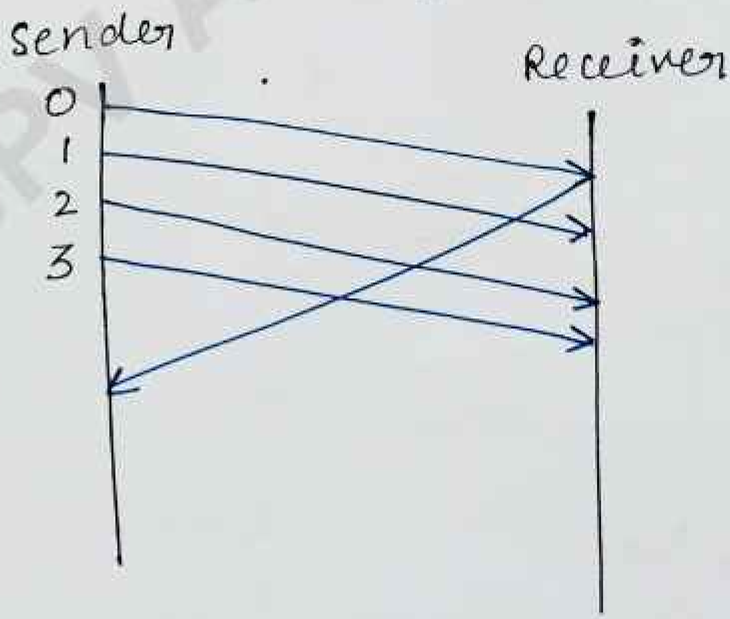
Sender will send first four frames to the receiver. (0, 1, 2, 3), N = 4



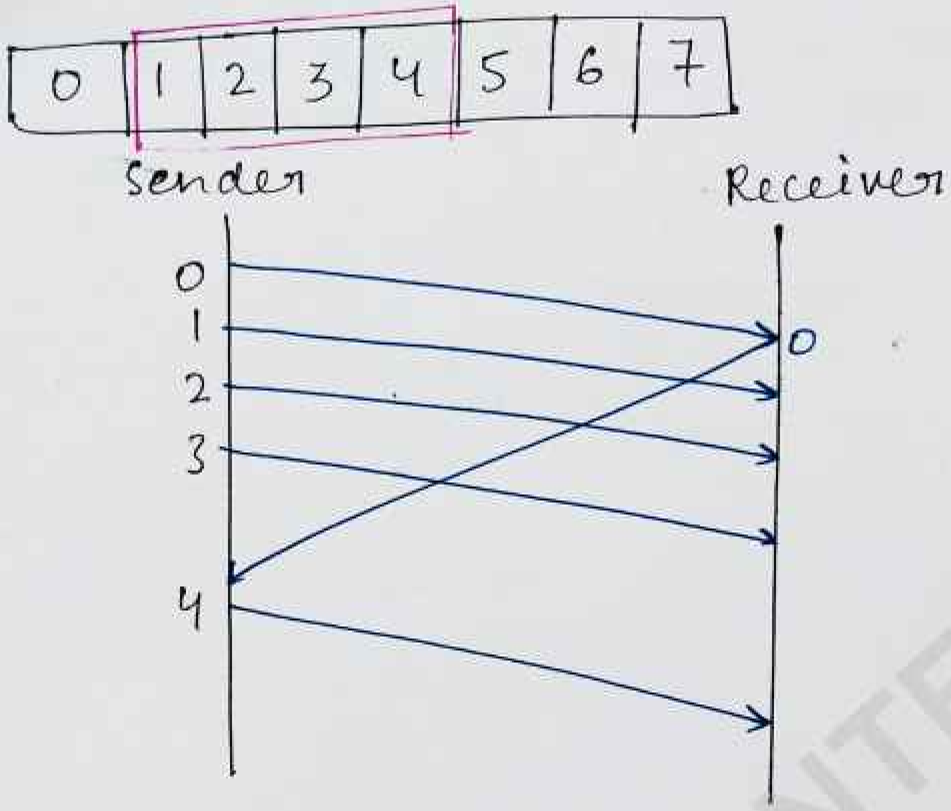


Step 2

Sender expects to get ack from 0th frame

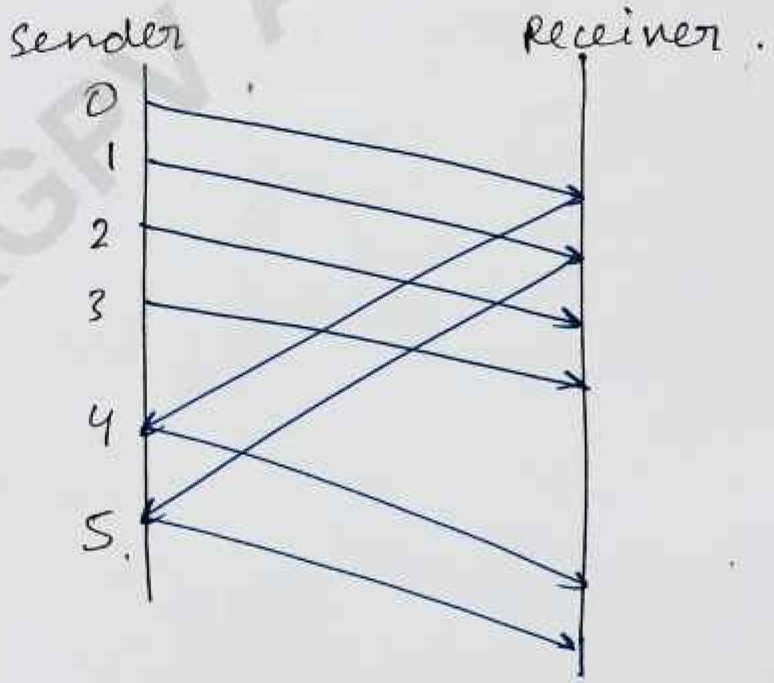


Step - 3



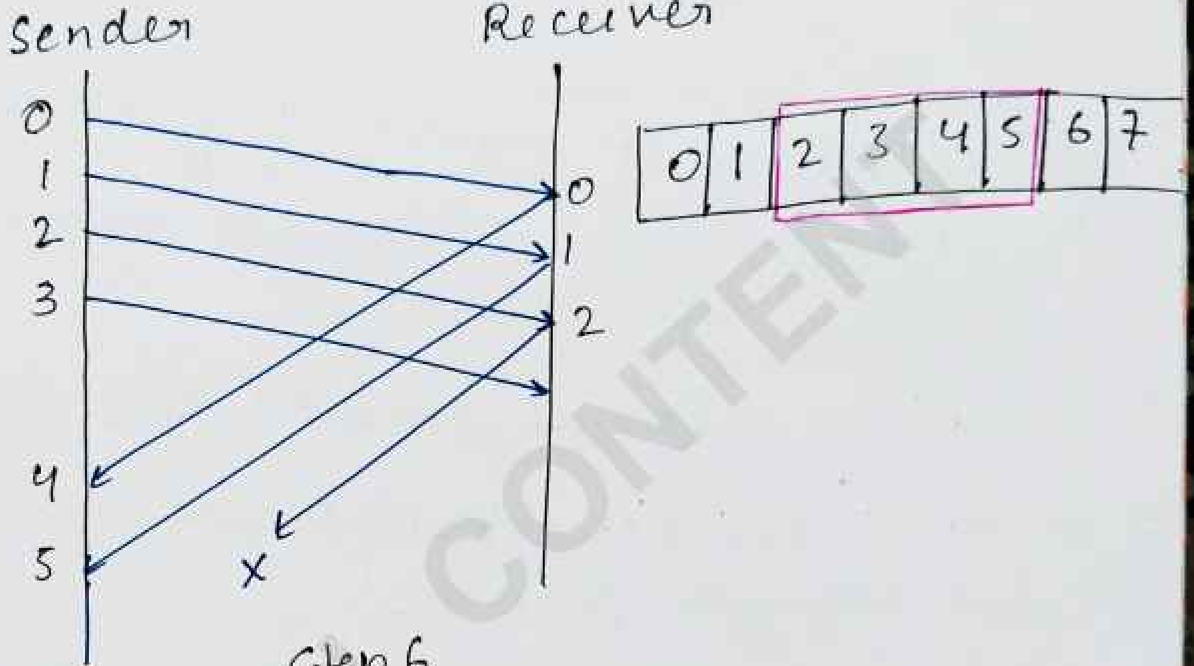
Step-4

sender get ack from 1th frame.



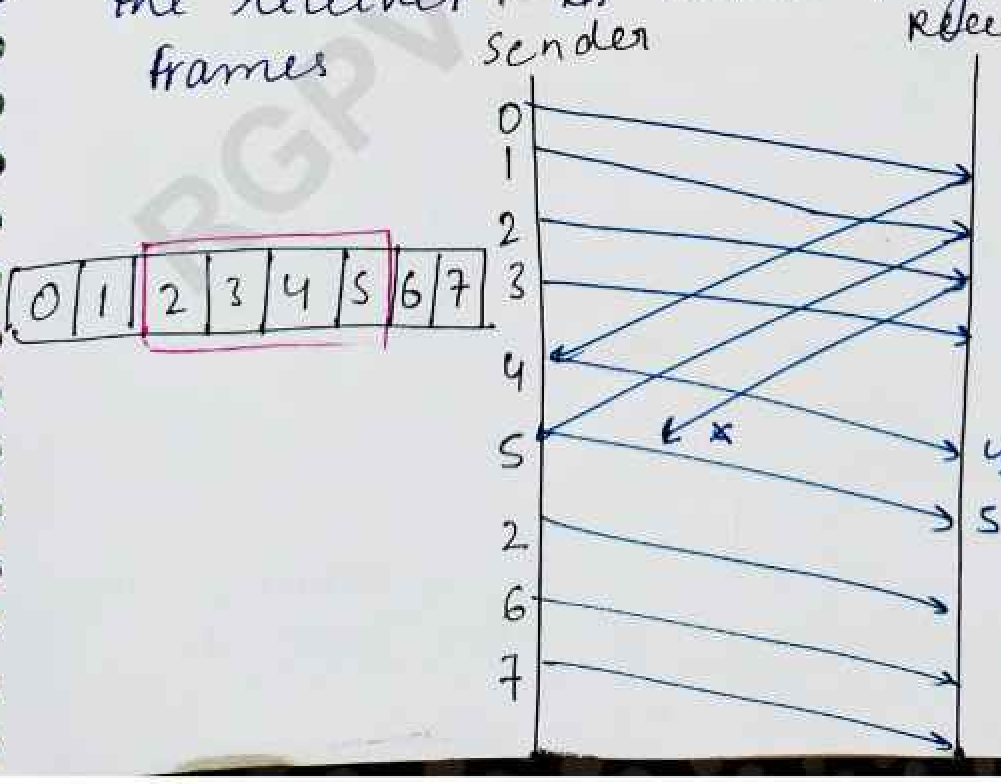
Step 5.

The ack for frame no. 2 will be sent by receiver. But ack is lost in network. Then receiver send Negative ack (NACK). of Frame 2 to the sender



Step 6

Sender will then send frame 2 alone to the receiver. As usual they send other frames



Protocol Verification

Protocol verification is the process of checking whether a communication protocol works correctly according to its design and rules. It ensures that data is transmitted properly between sender and receiver without errors, deadlocks or loss.

Main objectives of Protocol Verification.

1. Correctness - Checks if protocol follows all rules properly
2. Reliability → Ensures successful data delivery.
3. Deadlock Detection - Verifies that communication does not stop permanently.
4. Livelock Detection - Ensures system does not keep running without progress.
5. Error Handling - Confirms protocol can recovers from errors.

Finite State Machine (FSM) Model. 2023, 2024

This model are used for protocol verification and designing communication system.

FSM → A Finite State Machine is a Mathematical model used to represent a system with a finite number of states and transitions between those states based on inputs / events.

Components.

1. States - Different conditions of the system
2. Inputs - Events that cause change
3. Transitions - Movement from one state to another.
4. Output - Result after transition (optional)

Petri Net Model.

A Petri Net model is a graphical and mathematical tool used to represent and analyze the behavior of communication protocols in computer networks, especially systems with parallel activities.

It is widely used for protocol verification.

Components of Petri Net.

1. Places (Circles) → Represent states, conditions, or resources.

2. Transitions (Rectangles / Bars) → Represent events or actions that change the state

3. Tokens (Dots) → Represent the current status of the system.

4. Arcs (Arrows) connect places to transitions and transitions to places.

Use of Petri Net Model.

Petri Net is used to :-

- Data transmission protocols
- Flow control mechanism
- Resources sharing in networks
- Deadlock detection.
- Parallel communication systems

ARP 2022 (Most IMP)

★ ARP stands for Address Resolution Protocol.
It is used to find the MAC address of a device when its IP address is known

★ In a local network, communication happens using MAC addresses, but users usually known devices by IP addresses.

★ ARP helps to convert the IP address into a MAC address.

Working of ARP

1 A sender wants to send data to another device in the same network.

2 It knows the receiver's IP address but not MAC address.

3 Sender broadcasts an ARP Request:
"Who has this IP address?"

4 The device with that IP replies with an ARP Reply containing its MAC address.

5 Sender stores this mapping in the ARP cache for future use.

Example

→ If Device A wants to send data to IP 192.168.1.5.

It sends ARP request.

The device with IP 192.168.1.5 replies:

"My MAC Address is 00:1:2B:3C:40:5E"

ARP Packet contains

- Sender IP address
- Sender MAC address
- Target IP address
- Target MAC address

RARP

RARP stands for Reverse Address Resolution Protocol. It is a network used to obtain the IP address of a device when its MAC address is already known

It is the reverse of ARP

ARP → Finds MAC address from IP address.
RARP → Finds IP address from MAC address.

Why RARP is used →

→ When a computer starts, it may known its hardware (MAC) address, but not its IP address.

→ So the device sends a RARP request to a RARP server asking:

→ "This is my MAC address. Please tell me my IP address."

→ The server replies with the correct IP address.

Example → • MAC Address → 00-1A-2B-3C-4D-5E
• Device sends request to server.
• Server replies : IP Address = 192.168.1.10

GARP

GARP stands for Gratuitous Address Resolution Protocol or Gratuitous ARP.
It is a special type of ARP message in which a device sends an ARP request or reply for its own IP address.

It is used to announce or verify its IP-to-MAC address mapping on the local network

Why GARP used

GARP is mainly used for:

1. Updating ARP cache → Other devices update their ARP tables with the sender's new MAC / IP mapping.
2. Detecting Duplicate IP Address.
   checks whether another device is already using the same IP address.
3. Failover / High Availability.
   When a backup server becomes active, it sends GARP so traffic comes to the new MAC address.

Working of GARP

1. A device joins the network or changes interface status.
2. It sends a Gratuitous ARP packet saying: "My IP address is X
3. Other devices receives it and update their ARP cache.

Example

→ A computer with:

- IP Address → 192.168.1.20
- MAC Address → AA:BB:CC:DD:EE:FF

It sends GARP to inform all devices.

192.168.1.20 is at AA:BB:CC:DD:EE:FF